BORNEO hackwknd

FLEXD
Flex Your Ideas, Shape the Future

Cultura

AI-Powered Interactive Cultural Exploration

By Team FLEXD

A. INTRODUCTION

1. Background of Case Study

Over the past decade, the rapid advancement of Artificial Intelligence (AI) and interactive multimedia technologies has significantly transformed the paradigm of human–computer interaction. The relevance of this technological shift is reinforced by projections from Gartner (2022), which forecast a sharp global increase in the adoption of conversational AI, indicating a transition from passive visual interfaces toward two-way voice-based interaction systems.

This transformation aligns closely with the main theme of BorNEO HackWknd, "The Role of AI in ASEAN Social Impact," where AI is expected to be utilized not merely as a technological trend but as a meaningful tool for creating scalable solutions that improve educational access and empower communities across ASEAN. In response to this challenge, our team developed Cultura, an interactive cultural exploration experience that integrates Artificial Intelligence, real-time voice interaction, and immersive 3D characters to create a more engaging and participatory cultural learning environment. Within the ASEAN region home to an exceptionally diverse cultural heritage the preservation of indigenous cultures faces significant challenges in the digital era. This issue directly addresses the competition problem domains outlined in *Case Study 2: Indigenous Language Preservation* and *Case Study 3: AI for Cross-Culture Learning*. To tackle this widespread challenge, we conceptualized a scalable AI-driven platform capable of accommodating and preserving diverse ASEAN cultures. As a strong proof-of-concept and the first cultural module implemented within Cultura, we highlight the Minangkabau culture of Indonesia.

As an important cultural heritage of West Sumatra, Minangkabau culture carries significant educational value through its traditional customs, language, architecture, and performing arts. The philosophical principle "Adat Basandi Syarak, Syarak Basandi Kitabullah," the matrilineal kinship system, and heritage sites such as Istano Basa Pagaruyung represent valuable cultural assets that should be introduced to younger generations. However, the Atlas of the World's Languages in Danger report published by UNESCO, supported by youth demographic indicators from Badan Pusat Statistik (BPS), reveals that many local languages and cultural practices are experiencing a significant decline in everyday family use.

This decline represents a regional cultural crisis largely driven by the dominance of global digital media and the fact that cultural materials are still predominantly delivered through passive formats such as text and video. According to the Cone of Experience proposed by Edgar Dale and supported by modern e-learning research, learner memory retention typically ranges between 10%–20% when relying on passive methods, but can increase dramatically to 75%–80% through simulations and participatory interactive learning. These findings are further supported by studies conducted by Thomas M. Connolly et al. (2012) and James Paul Gee (2003), which demonstrate that digital interactive environments significantly enhance learning motivation and engagement.

As Software Engineering students, we recognize the urgency of empowering indigenous communities while promoting digital inclusion through technological innovation. Therefore, we engineered a real-time voice-interactive Virtual 3D Assistant, which serves as the core technological component of the cultural exploration experience within Cultura. The system architecture captures voice input from users and processes it through a conversational AI platform such as Convai using three primary pipelines: Speech-to-Text (STT), Natural Language Processing (NLP), and Text-to-Speech (TTS). Due to its modular architecture, the NLP system can be trained with cultural datasets from any ASEAN region. For the current prototype within Cultura, the system is specifically trained using a Minangkabau cultural knowledge base.

This data is then seamlessly integrated into a game engine such as Unity to render real-time lip-sync animations and behavioral gestures for the 3D character. This approach transforms the 3D character into an intelligent cultural guide capable of bringing indigenous language and cultural knowledge to life through immersive interaction. Ultimately, Cultura lays the technological foundation for a broader cross-cultural preservation movement across ASEAN, leveraging AI to make cultural education more engaging, accessible, and sustainable in the digital era.

2. Chosen SDG

In accordance with the BorNEO HackWknd guidelines, this innovative solution primarily targets SDG 4: Quality Education by leveraging AI to enhance the accessibility, inclusivity, and effectiveness of learning about diverse indigenous heritages across the ASEAN region. While the Minangkabau culture serves as our initial proof-of-concept and currently available

feature, our platform is engineered to be highly scalable to educate users about any ASEAN culture that has historically been difficult to access interactively. Secondarily, this solution supports SDG 11: Sustainable Cities & Communities by providing adaptable, community-centred tools to safeguard, digitize, and promote local cultural heritage and identity. By establishing a robust technological framework capable of accommodating various cultural datasets, we ensure that indigenous identities from Minangkabau to the wider ASEAN communities remain profoundly relevant and sustainable amidst rapid urban modernization.

3. Problem Statement

Current passive learning methods fail to engage digital natives, causing a critical decline in the preservation of diverse indigenous cultures across the ASEAN region. This widespread crisis directly hinders SDG 4 (Quality Education) and SDG 11 (Sustainable Cities & Communities). To combat this, a scalable, cross-cultural technological framework is urgently needed. While our current platform utilizes the Minangkabau culture as its initial proof-of-concept and inaugural feature, the system is engineered to be adaptable for any ASEAN heritage. Based on this vision, the specific pain points and core technical barriers our project aims to solve are:

1. The lack of an optimal, scalable design for real-time voice interaction systems on 3D characters using the Convai platform for cultural education.

2. The technical difficulty in seamlessly integrating Convai with game engines to enable the virtual character to respond to the user's voice naturally.

3. The absence of empirical data regarding user responses to 3D characters equipped with real-time voice interaction.

4. The unmeasured extent to which such developed systems can actually increase user engagement in interactive media compared to traditional passive methods.

4. Objectives

To solve the identified problems and achieve a meaningful, scalable ASEAN Social Impact, the specific goals of our solution are:

1. To design and develop a highly adaptable, real-time voice-interactive Virtual 3D Assistant system using the Convai platform as a scalable educational framework for diverse ASEAN cultures, utilizing the Minangkabau culture as our inaugural proof-of-concept and currently available feature.

2. To seamlessly integrate speech recognition technology, conversational AI, and character animation to produce natural, responsive two-way communication that is capable of adapting to various local cultural and linguistic contexts.

3. To empirically evaluate user responses towards the developed 3D character system, validating its effectiveness as a standardized technological blueprint for broader cross-cultural learning and indigenous language preservation.

4. To prove that the implementation of this AI-driven interactive system significantly increases user engagement and enhances the overall learning experience compared to passive media, establishing a strong foundation for future expansion to other indigenous communities across the ASEAN region.

B. REVIEW OF EXISTING SOLUTIONS

1. Analyze current solutions or competitors in the market

Research on real-time voice interaction for 3D characters and AI-based conversational agents has grown rapidly over the past decade. However, current state-of-the-art solutions still exhibit fragmented focuses and lack adaptability for regional cultural education. Several relevant scientific publications outline the current landscape:

a. 3D Virtual Humans and Multimodal Interaction Design: Wang (2024) conducted a comprehensive study on the interaction design of virtual humans, highlighting speech as a primary modality. The study emphasizes that design must combine speech with facial expressions and gestures to enhance User Experience (UX).
   - *Achievement*: Increased learning engagement through conversational agents.

- *Limitation*: Lacks the effective integration of 3D characters with real-time speech interaction.

b. AI-Enhanced Conversational Avatars for Immersive Interaction: A review in the *World Journal of Advanced Research and Reviews* (2025) examines AI avatars integrating NLP, Speech Recognition, and multimodal cues. It found that improved interaction relies on merging AI components with responsive character design, though ethical and bias risks remain.
   - *Achievement:* Successful integration of expressions and conversation in virtual humans.
   - *Limitation:* The system does not focus on integrating a real-time speech pipeline within a game engine environment.

c. LLM-Driven 3D Hyper-Realistic Interactive Digital Humans: Song & Xiong (2025) introduced a digital human system utilizing ASR, NLP, and emotional TTS within a modular architecture. It demonstrated the ability to manage complex conversations and maintain context using Large Language Models (LLMs).
   - *Achievement:* Significant improvement in NLP quality and AI dialogue realism.
   - *Limitation:* Has yet to fully integrate speech interaction seamlessly with 3D characters.

d. ViBES (Voice-Integrated Body and Expression System): Zhang et al. (2025) introduced ViBES, a 3D agent model that simultaneously generates body movements and facial expressions synchronized with dialogue, creating a more natural, real-time social interaction.
   - *Achievement:* High-quality real-time facial animation and expression synchronization.
   - *Limitation:* Does not integrate voice-based conversational AI.

e. High-Fidelity Real-Time Conversational Digital Humans: Huang et al. (2025) developed 3D digital humans with an asynchronous execution pipeline to optimize real-time responses, including wake-word detection and emotional prosody in TTS.
   - *Achievement:* Voice interaction significantly increases user trust.
   - *Limitation:* Lacks the integration of modern conversational AI with real-time 3D characters.

2. Highlight gaps or limitations that your project aims to fix

Based on the review of previous research, while virtual humans have advanced significantly in NLP and visual expressions, most studies develop these technological components in silos. The core limitations our project aims to solve are:

a. Lack of Real-Time Integration within Game Engines: Most existing research develops AI conversational systems separately without direct implementation in interactive multimedia environments like game engines. Our Fix: We seamlessly integrate the Convai platform directly into Unity 3D, enabling real-time, highly responsive audio-visual synchronization.
b. Absence of Scalable AI in Local Cultural Education: Current solutions overwhelmingly focus on generic applications (e.g., customer service, generic social avatars), while applications for indigenous cultural preservation are incredibly scarce. Our Fix: We engineered a scalable, modular educational framework. While our current inaugural feature uniquely utilizes a Minangkabau cultural dataset as a proof-of-concept, the architecture is designed to easily ingest and preserve diverse cultural datasets from across the ASEAN region.
c. Limitations in Immersive and Participatory Experiences: Many digital learning systems remain passive, failing to generate high user engagement among digital natives. Our Fix: We shift the paradigm from passive reading/watching to an immersive, two-way conversational simulation proven to boost learning retention.
d. Lack of Culturally Contextual 3D AI Media: There is a critical gap in research integrating conversational AI with 3D learning media that is deeply contextualized to local heritage. Our Fix: Our Virtual 3D Assistant is not just a chatbot; it is designed as an adaptive, culturally accurate digital guide, laying the technological blueprint for cross-cultural learning throughout ASEAN.

C. METHODOLOGY

1. Technical Track

To achieve our vision of a cross-cultural preservation platform across ASEAN, the development of the Virtual 3D Assistant integrates interactive multimedia technology, Conversational Artificial Intelligence, and real-time voice processing into a single, cohesive system architecture. This approach aims to create an immersive interaction experience where

a 3D virtual character communicates naturally, acting as a highly intelligent human guide. The system is designed to enable two-way, voice-based communication through real-time natural language processing. By leveraging a combination of speech recognition, NLP, conversational AI, and text-to-speech, the virtual character can understand user queries and provide contextual, informative responses. This methodology is highly relevant for developing a scalable, culture-based educational framework because it presents information interactively and adaptively, making it easily understandable for users from diverse backgrounds across the ASEAN region.

Technology Stack: The development utilizes several core, highly integrated technologies to support real-time audio-visual interaction. These include:

1. Game Engine: Unity 3D serves as the primary engine to build the interactive multimedia application. Unity was selected for its robust capability in handling 3D graphics, character animation systems, and flexible external API integrations required for AI-based development.
2. Conversational AI Platform: Convai is utilized as the core conversational intelligence engine, enabling the virtual character to understand and respond to user speech naturally. Convai provides dialog management, context processing, and direct integration with 3D characters within the Unity environment.
3. Speech Recognition (STT): This technology converts the user's voice input into digital text that the AI system can process, serving as the initial trigger in the voice interaction pipeline.
4. Natural Language Processing (NLP): The NLP module analyzes the transcribed text to understand the intent, context, and linguistic structure of the user's query. Because our system is modular, this NLP engine can process any language or cultural context.
5. Text-to-Speech (TTS): This technology converts the text responses generated by the AI back into an audible voice, allowing the 3D character to answer verbally, making the interaction feel highly natural and communicative.
6. 3D Character System: The 3D character is developed using a digital creation pipeline that includes modeling, rigging, and expression animation. It is then integrated into Unity and linked to the AI system, acting as an interactive agent that responds via voice, facial expressions, and behavioral gestures.

System Architecture: The system architecture is designed around an AI-driven, real-time voice interaction pipeline. This continuous loop consists of the following main stages:

1. Voice Input: The interaction initiates when the user asks a question or gives a command via a microphone.

2. Speech Recognition: The received audio signal is immediately processed by the speech recognition module and converted into analyzable digital text.

3. Natural Language Processing: The generated text is processed using the NLP module to identify user intent and deeply understand the conversational context.

4. Conversational AI Processing: The processed query is routed to the Convai platform. The AI analyzes the question against a configured cultural knowledge base. *Crucially, while this knowledge base currently utilizes Minangkabau cultural data as our inaugural proof-of-concept, the modular architecture allows it to easily ingest datasets from any other ASEAN culture.* The AI then generates a relevant and informative dialog response.

5. Text-to-Speech Output: The generated text response is converted into audio using TTS technology.

6. Character Animation Response: In the final stage, the 3D character delivers the response through a synchronized combination of audio, lip-sync, body movements, and facial expressions, successfully creating an immersive experience akin to conversing with a real cultural guide.

Implementation Process: The implementation is executed iteratively to ensure optimal technological integration and low latency. The development phases include:

1. Integrating the 3D character into the Unity environment as the visual representation of the virtual assistant.

2. Configuring the Convai platform as the conversational AI engine that manages the interactive dialog logic.

3. Executing API integration between Convai and Unity, directly connecting the 3D character to the AI conversation system.

4. Developing the modular knowledge base and dialog scenarios. Currently, this focuses exclusively on Minangkabau cultural information (traditions, history, and values) to serve as a robust proof-of-concept, structured so that other ASEAN cultural modules can be seamlessly plugged in later.

5. Implementing the real-time voice interaction pipeline that seamlessly connects the speech recognition, NLP, dialog AI, and text-to-speech modules.

6. Conducting iterative system testing to ensure communication stability, dialog accuracy, and perfect synchronization between the generated voice and character animation.

D. RESULTS

1. Summary Of Work Results

We successfully developed and implemented the Virtual 3D Assistant system powered by conversational AI, fully meeting our core objectives. As an inaugural proof-of-concept, the system effectively interacts with users via real-time voice communication to educate them about the Minangkabau culture. The developed 3D character is fully capable of:

a. Accurately understanding user queries through Speech Recognition (STT).
b. Processing conversational context and cultural meaning using modular NLP technology.
c. Delivering highly relevant, synchronized audio responses via Text-to-Speech (TTS).

This seamless integration produces a significantly more natural, interactive, and immersive experience compared to conventional digital learning media that rely solely on passive text or video. Ultimately, it proves that our scalable architecture can effectively deliver complex cultural information such as history, customs, and traditional values through dynamic, human-like dialogue.

2. Data Or Feedback

Based on initial prototype testing and user feedback gathered during development, digital natives exhibited an overwhelmingly positive response to the real-time voice interaction concept. Users reported that the system provided a far more captivating learning experience compared to traditional passive methods. The conversational interface allowed users to acquire cultural knowledge naturally, as if conversing with a real human cultural guide. This strongly

indicates that our conversational AI approach drastically boosts user engagement in digital learning. Furthermore, technical evaluations confirmed that the integration of speech recognition and text-to-speech technologies operates stably within the application environment, facilitating smooth, low-latency, real-time two-way communication between the user and the virtual character.

3. Social Impact

The development of this Conversational AI-driven Virtual 3D Assistant yields a profound social impact, perfectly aligning with ASEAN's vision for digital education, equitable access to knowledge, and the preservation of indigenous heritage. By converging Artificial Intelligence, natural language processing, and interactive multimedia, this system is engineered to create a highly inclusive, engaging, and relevant learning experience tailored for the digital generation.

a. **Impact on Digital Education** In the educational sector, this platform serves as a breakthrough alternative learning medium that significantly boosts user involvement. Unlike conventional one-way text or video methods, our AI-driven conversational approach empowers users to actively participate through direct dialogue with the virtual character. Through the seamless integration of STT, NLP, and TTS, users can ask spontaneous questions, seek detailed explanations, and explore cultural data flexibly and personally. This fosters an immersive, adaptive, and highly contextual learning environment, which drastically improves comprehension and stimulates the user's interest in cultural studies. The system is ready to be deployed in schools, educational museums, and broader tech-based learning platforms.

b. **Expanding Access to Cultural Knowledge** A primary barrier in cultural education is the limited access to authentic, interactive informational sources, which are often confined to physical museums, cultural centers, or formal institutions. Through this AI-powered Virtual 3D Assistant platform, rich cultural information can be accessed digitally by society from any region, breaking down geographical dependencies. This allows the system to reach users who previously faced barriers in accessing cultural learning resources. Consequently, this technology contributes to creating a more inclusive and equitable distribution of knowledge, ensuring that people from diverse backgrounds can easily study and understand local cultures.

c. **Contribution to Cultural Preservation** From a preservation standpoint, the approach utilized in this project pioneers a new frontier in the digitization of intangible cultural heritage. Complex information regarding history, traditions, social values, and cultural philosophies currently prototyped with the Minangkabau culture is packaged into a highly attractive interactive medium for the youth. Transforming ancestral knowledge into a 3D AI-driven virtual assistant makes cultural dissemination highly communicative and deeply relevant to modern technological lifestyles. This successfully bridges the generational gap, significantly increasing the youth's desire to recognize, understand, and cherish indigenous values, ensuring traditional heritage is broadly documented and studied within a modern educational ecosystem.

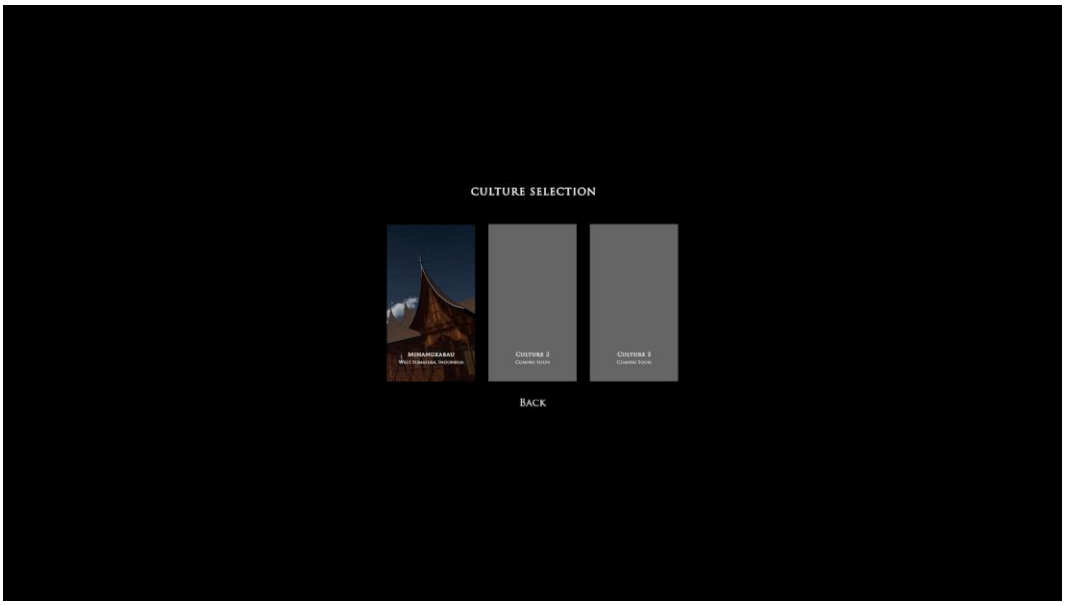
d. **Development Potential and Scalability** Long-term, this Conversational AI-based Virtual 3D Assistant is fundamentally engineered as a highly scalable cultural education platform. While currently demonstrating success with Minangkabau data, the modular architecture can be rapidly expanded to encapsulate various other local cultures across Indonesia and the wider ASEAN region. Using the exact same technological approach, diverse virtual characters can be developed and trained to represent the unique heritages of different regions, ultimately weaving a comprehensive, integrated digital cultural ecosystem. Therefore, this technology is not merely an innovation in digital education; it is a strategic, scalable solution to drive cultural digitization, fortify local identities, and elevate public awareness of ASEAN's rich cultural diversity in the digital transformation era.

E. PROTOTYPE SHOWCASE

1. Output / UI Screenshots

The following UI screenshots illustrate our scalable, cross-cultural technological framework in action. While the platform is fundamentally designed to house diverse ASEAN cultural datasets, these images highlight our inaugural Minangkabau cultural module as the primary proof-of-concept.

The **Main Menu interface** presents two primary options: **"Start"** and **"Exit."** The background features **Rumah Gadang**, the traditional house of the Minangkabau people from West Sumatra, Indonesia. This visual element is intentionally used to immediately introduce the cultural identity that forms the central theme of the interactive experience.



The **Culture Selection Menu** allows users to select a cultural experience to explore. Currently, the system includes one available cultural module, namely **Minangkabau culture from Indonesia**. The system architecture is designed modularly, enabling the integration of additional cultural content in future developments.

The initial interaction scene introduces the virtual character **Datuak Rajo Alam**, who serves as a cultural guide powered by **Conversational Artificial Intelligence**. In this scene, the character introduces himself to the user and begins guiding them through various aspects of Minangkabau culture. The background features a **Rumah Gadang**, representing the cultural environment being explored.

The **interactive dialogue interface** illustrates a conversation between the user and **Datuak Rajo Alam**. In this interaction, the user asks about traditional Minangkabau cuisine. The character responds based on the cultural knowledge base integrated into the AI system.

The **Pause Menu interface** provides several interaction control options: **"Resume," "Restart," and "Quit."** This menu allows users to continue the experience, restart the exploration session, or exit the system.

This scene shows **Datuak Rajo Alam** walking while inviting the user to follow him during the cultural exploration inside the **Rumah Gadang**. This interaction is designed to create a more immersive experience by presenting a virtual guide who actively leads the user through the cultural environment.

This scene shows **Datuak Rajo Alam** explaining the architectural structure of the **Rumah Gadang** to the user. The explanation includes key elements of Minangkabau traditional architecture based on the cultural knowledge embedded in the AI system.

This scene shows **Datuak Rajo Alam** explaining the function and cultural meaning of the **"Bilik,"** which refers to the private rooms inside the **Rumah Gadang**. The explanation highlights how each part of the traditional Minangkabau house carries specific cultural values and social functions within the community.

F. AI ACKNOWLEDGEMENT (MANDATORY)

1. **Disclosure:** We transparently disclose the use of AI tools in the development and documentation of this project.
2. **Usage:** In developing this scalable ASEAN cross-cultural platform, we utilized several AI-driven tools to accelerate development, refine our ideation, and power the core application.

The specific AI tools and their usages are as follows:

a. Convai

Integrated as the primary Conversational AI platform to power the system's core feature. It was used to handle the Speech-to-Text (STT), Natural Language Processing (NLP), and Text-to-Speech (TTS) pipelines, specifically managing our modular cultural knowledge base (starting with our Minangkabau inaugural proof-of-concept) to enable the 3D character to understand and respond to users in real-time.

b. Gemini

Utilized for linguistic refinement, translation, and compiling the final structure of this hackathon report to ensure a highly professional, bilingual presentation.

c. ChatGPT

Utilized during the initial brainstorming phases to assist in idea exploration, formulating early research concepts, and refining the structural logic of the project's foundational framework.

Final Disclaimer: All Artificial Intelligence (AI) technologies in this project are utilized both as supporting tools during the development and report preparation process, and as the core component that powers the system’s conversational interaction capabilities. Meanwhile, the core concept design, cultural data curation, system architecture planning, user experience (UX) design, and final system evaluation were entirely designed and carried out independently by our development team.

G. CONCLUSION

This project successfully designed and developed a highly scalable, AI-driven Virtual 3D Assistant framework, utilizing the Minangkabau culture as our successful inaugural proof-of-concept. By seamlessly integrating Speech Recognition (STT) to capture user input, Natural Language Processing (NLP) to understand conversational context, and Text-to-Speech (TTS) to generate natural audio responses, the 3D character effectively functions as a real-time, interactive cultural agent within a digital multimedia environment.

The outcomes of this development prove that shifting from passive text and video (which traditionally yield low memory retention) to an immersive, dialogue-based approach creates a highly participatory learning experience. Through intuitive two-way conversations, digital natives can easily absorb complex historical facts, traditional customs, and cultural values. This demonstrates the massive potential of Conversational AI to shatter conventional boundaries, significantly boosting user engagement and enriching the overall digital education experience.

Beyond a single cultural application, this research establishes a robust technological blueprint engineered for massive scalability across the ASEAN region. The developed Virtual 3D Assistant can be broadly deployed across digital museums, interactive e-learning platforms, educational simulations, and culture-based educational games. Consequently, this technology serves not merely as a learning tool, but as an innovative, region-wide infrastructure to support the digitization and preservation of indigenous identities in the digital transformation era.

Despite these successes, we recognize several limitations that present exciting opportunities for future improvements. On the technical front, future development will focus on enhancing multimodal interaction capabilities.

This includes deepening emotional synchronization, integrating advanced facial expressions, and adding dynamic body gestures so the virtual character's responses feel even more realistic and human-like. Furthermore, the modular NLP system will be expanded to comprehensively support specific local dialects (starting with the Minangkabau language and scaling to other ASEAN indigenous languages) to ensure maximum authenticity and cultural context.

Finally, for future research, we strongly recommend conducting large-scale, cross-regional user evaluations utilizing comprehensive measurement methods. This will allow us to

continually assess the system's definitive impact on user engagement, comprehension, and long-term knowledge retention. Through these continuous improvements, we envision this Conversational AI-driven Virtual 3D Assistant becoming a landmark ed-tech innovation, spearheading the movement for AI-based learning and cross-cultural preservation throughout the ASEAN region.

BIBLIOGRAPHY

1. Connolly, T. M., Boyle, E. A., MacArthur, E., Hainey, T., & Boyle, J. M. (2012). An update to the systematic literature review of empirical evidence of the impacts and outcomes of digital games on learning. Computers & Education, 59(2), 1449–1461. https://doi.org/10.1016/j.compedu.2012.01.004
2. Gee, J. P. (2003). *What video games have to teach us about learning and literacy*. Palgrave Macmillan.https://eric.ed.gov/?id=ED512742
3. Dale, E. (1946). *Audio-visual methods in teaching*. Dryden Press. https://elearningindustry.com/cone-of-experience-what-really-is
4. Badan Pusat Statistik. (n.d.). *Youth demographic indicators*.https://www.bps.go.id
5. Gartner. (2022). *Projections on conversational AI adoption*.https://www.gartner.com
6. UNESCO. (n.d.). *Atlas of the world's languages in danger*.https://ich.unesco.org/en/RL/atlas-of-the-worlds-languages-in-danger-00223
7. Følstad A, Brandtzaeg PB. Chatbots and the new world of HCI. Interactions.
8. 2017;24(4):38–42.
9. Winkler R, Söllner M. Unleashing the potential of chatbots in education: A state-of-the
10. art analysis. Bus Inf Syst Eng. 2018;60(4):345–353.
11. Zhou L, Gao J, Li D, Shum HY. The design and implementation of conversational AI
12. chatbots. Artif Intell Rev. 2020;53:4573–4595.
13. Hussain S, Ginige JA, Richards D. Conversational agents for education: A systematic
14. literature review. Educ Inf Technol. 2021;26:1–33.
15. Kocaballi AB, Berkovsky S, Quiroz JC, Laranjo L, Tong HL, Rezazadegan D, et al. The
16. personalization of conversational agents in healthcare: Systematic review. J Med Internet
17. Su J, Yang W. Unlocking the power of ChatGPT: A framework for applying generative AI
18. in education. Educ Inf Technol. 2023;28:1–18.
19. Huang X, Zhang Y, Li Z, Liu W. High-fidelity real-time conversational digital humans
20. with emotional speech synthesis. IEEE Trans Vis Comput Graph. 2025;31(2):1–12.
21. Zhang H, Song Y, Liu T, Wang X. ViBES: Voice-integrated embodied agents for real-time